Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18DS7DE51]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

NATURAL LANGUAGE PROCESSING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. List and explain various text preprocessing techniques in NLP with example. **10**

b. Consider the given text "English is a natural language; Java is a programming language. NLP enables clearer human-to-machine communication, without the need for the human to speak". **04**

What will be outcome of the text after applying the following tasks on it.

i. Tokenization

ii. Stopword removal

2. a. Distinguish between the syntactic and semantic analysis in NLP. **06**

b. Discuss the important roles of tokenisation in lexical analysis with example. **08**

3. a. Define language generation. Give the simple language generation steps in NLP. **06**

b. Explain the text planning with its types of approaches in NLP. **08**

OR

4. a. Write in detail about the Text planning steps applied in NLG. **10**

b. Write a python program for a text generation using demo_grammar library. List the grammar supported by demo_grammar library. Generate manual output for generating 4 different combinations of sentences using given Grammar. **04**

5. a. Explain the following concepts: **08**

i. Corpus balance

ii. Sampling

iii. Representativeness

iv. Corpus size

b. Annotate the following text by describing the suitable PoS tags. **06**

"Robert was a good king. He had a great army. He wanted to bring peace to his kingdom".

OR

6. a. Write in detail about the process of building the Tree bank. **10**

b. **04**

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

In the above sample data set, explain about the type of each attribute such as categorical, numeric and also which are the attributes can be used as input and target attributes while training phase.

7. a. Explain supervised, unsupervised and reinforcement learning methods with relevant examples. **09**

b. Differentiate between generative and discriminative models in NLP. **05**

OR

8. a. Write in detail about the statistical parsing with suitable examples? **09**

b. What language can be used to create a body corpus in NLP, explain with example? **05**